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An estimation model for determining the annual energy cost budget in educational facilities using SARIMA (seasonal autoregressive integrated moving average) and ANN (artificial neural network)

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ABSTRACT

Electricity consumption in educational facilities has increased by an annual average of 9.84% since 2006. However, it is not considered as a factor in determining the AECB (annual energy cost budget) in South Korea. Therefore, this study aims to develop an estimation model for determining the AECB in educational facilities using the SARIMA (seasonal autoregressive integrated moving average) model and the ANN (artificial neural network). This study collected electricity consumption data for 7 years (2005–2011) from 787 educational facilities. The result of this study showed that the prediction accuracy of the proposed hybrid model (which was developed by combining SARIMA and ANN) was improved, compared to the conventional SARIMA model. The MAPE (mean absolute percentage error) of the proposed method and conventional method for determining the AECB in educational facilities was determined at 0.11–0.24% and 1.23–1.84%, respectively. Namely, it was determined that the proposed method was superior to the conventional method. The proposed model could enable executives and managers in charge of budget planning to accurately determine the AECB in educational facilities. It could be also applied to other types of resources (e.g., water consumption or gas consumption) used in educational facilities.

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1. Introduction

As of 2009, South Korea is ranked as the 10th largest country in final energy consumption (229 Mtoe), the 9th in electricity consumption (437 TWh), and the 8th in CO_2 emissions from energy combustion (518.1 Mt CO_2) [1–4]. Particularly, in the last five years (2006–2011), electricity consumption in Seoul, the capital of South Korea, has increased by about 19%. Also, after an incident on September 15, 2011, which almost caused a blackout, the entire country reconsidered the importance of predicting electricity consumption [5]. Meanwhile, as of 2011, 58.2% of the total energy consumption in Seoul has been from construction industry [6]. Accordingly, Seoul introduced the *Total Amount of Energy Consumption*, and established a plan to reduce building energy consumption in 2030 by 20% compared to 2000 [7,8].

In 2011, a total of 1282 educational facilities (elementary school: 591; middle school: 377; and high school: 314) have been operated in Seoul, consuming about 608,055 MWh of electricity [9]. This is about 17.18% of the electricity consumption of the public sector building in Seoul [10,11]. Thus, educational facilities must save electricity consumption to achieve Seoul's energy saving goal. Also, in the last five years (2006–2011), the electricity consumption of educational facilities in Seoul has increased by an annual average of 12.59% (elementary school: 14.17%; middle school: 11.40%; and high school: 12.20%). This is due to the increase in electricity consumption caused by electronic appliances and electric heat pump systems that were recently introduced in educational facilities [12–14]. Such a rapid increase in electricity consumption has caused the following two problems. First, a shortage of electricity supply for heating in the winter and cooling in the summer has resulted from a lack of the AECB (annual energy cost budget) in educational facilities. Accordingly, the quality of the educational environment has been degraded. Second, the increase in the ratio of the AECB in educational facilities has caused the lack of a budget for the other purposes. Accordingly, educational facilities have used the unbalanced budget for the building operation and maintenance.

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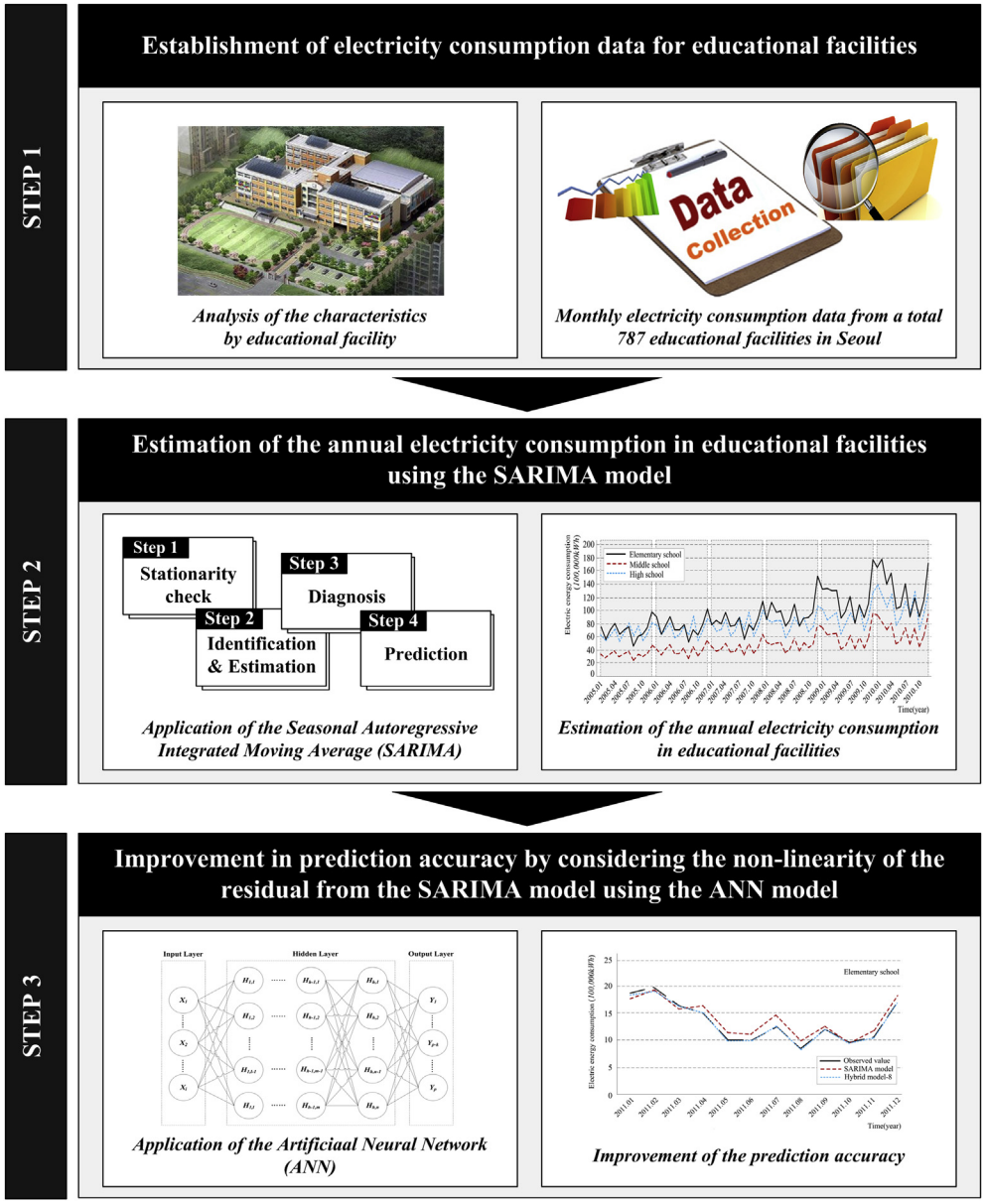


Fig. 1. Research framework.

Previous research implemented two methods for time series forecasting of energy consumption: (i) linear approach: MRA (multiple regression analysis); ARIMA (autoregressive integrated moving average) model; and SARIMA (seasonal autoregressive integrated moving average) model; and (ii) non-linear approach: ANN (artificial neural network).

First, the linear approach is suitable for cases where independent and dependent variables are in a linear relationship. While the MRA model is simple and efficient, its prediction accuracy is low [16–18]. Mohamed and Bodger [19] and Baker and Rylatt [20] attempted to predict electricity consumption in New Zealand and the UK domestic energy demand, respectively, using the MRA model. The ARIMA model, a methodology developed by Box and Jenkins, can predict future values using the auto-correlation analysis of time series data [21]. If there is a seasonal pattern in time series data, the SARIMA model can be used [22]. Gonzales et al. [23] predicted the future values of five energetic variables (i.e., domestic electricity consumption, industrial electric energy consumption, black-coal production, anthracite production, and electric energy production) in Northern Spain. Compared to exponential smoothing and the MRA model, it was determined that the ARIMA model was the most superior.

Second, the non-linear approach is suitable for cases where independent and dependent variables are in a non-linear relationship. Previous research used the ANN model, a representative methodology in non-linear approaches. Geem and Roper [24] predicted energy demand in South Korea based on four variables (i.e., GDP (gross domestic production), population, import, and export amounts). Compared to exponential smoothing and the MRA model, the ANN model was shown as the most superior. Ekonomou [25] predicted energy consumption using four variables (i.e., yearly ambient temperature, installed power capacity, yearly per resident electricity consumption, and GDP). Compared to the MRA model and the support vector machine, the ANN model was shown as the most superior. Zhang and Qi [26] developed an ANN model considering seasonality, pointing out

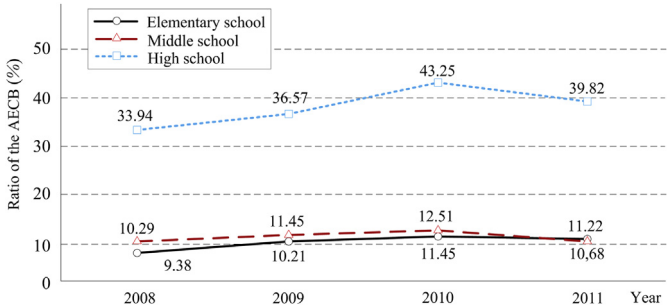


Fig. 2. Ratio of the AECB to the common operational cost (2008–2011).

the need for effective data pre-processing to reflect seasonal or trend variation.

However, actual problems usually have both linear and non-linear characteristics, and previous research attempted to explain the variability of actual problems by combining two methods [27–37]. Koutroumanidis et al. [35] predicted fuel wood prices in Greece through a hybrid model that combined the ARIMA and the ANN models. Tseng et al. [36] predicted the total production of the machinery industry and the soft drink industry in Taiwan through a hybrid forecasting model, known as SARIMABP model that combined the SARIMA model and neural network BP (back propagation) model. Shafie-khah et al. [37] predicted energy price using ARIMA, radial basis function neural networks, and partial swarm optimization.

Previous research attempted to improve the prediction accuracy of future values by combining linear and non-linear approaches. Therefore, this study aims to develop an estimation model for determining the AECB in educational facilities using the SARIMA model (linear approach) and the ANN model (non-linear approach). This study was conducted in three steps: (i) establishment of electricity consumption data for educational facilities; (ii) estimation of the annual electricity consumption in educational facilities using the SARIMA model; and (iii) improvement in prediction accuracy by considering the non-linearity of the residual from the SARIMA model using the ANN model (refer to Fig. 1).

2. Current status of the AECB in educational facilities

2.1. Trends in the AECB in educational facilities in Seoul

The SMOE (Seoul Metropolitan Office of Education) established the *Guideline for the School Budget* to guarantee the autonomy of the educational facility in budget operation. According to this guideline, the common operational cost (a dependent variable) is calculated using a linear equation with independent variables such as the number of students, the number of classes, the total floor area, and the elapsed year. However, despite the fact that the common operational cost includes the AECB, electricity consumption is not considered in the process of establishing the common operational cost [38].

Fig. 2 shows the ratio of the AECB to the common operational cost in the last four years (2008–2011). In general, the ratio of the AECB tends to increase each year [39–42]. The result shows that, in the process of establishing the common operational cost, the trend in electricity consumption could not be sufficiently considered. Accordingly, the shortcomings of the AECB could cause insufficient operation of heating and cooling systems, which ultimately degrade the quality of the educational environment. Also, the increase in the ratio of the AECB in educational facilities could cause the lack of a budget for the other purposes, which ultimately resulted in the unbalanced budget for the building operation and maintenance [15].

2.2. Conventional method for determining the AECB in educational facilities

According to interviews with executives and managers in charge of budget planning in SMOE, the AECB in educational facilities is established based on the electricity consumption of the previous year. For example, the AECB in 2011 is established based on the AECB in 2010. If this conventional method is used, however, the trend in the increase in electricity consumption cannot be considered. Therefore, it is necessary to develop a methodology that is capable of accurately predicting the trend in the increase of electricity consumption in educational facilities, and establishing the AECB accurately based on such prediction.

3. Materials and methods

3.1. Establishment of data set

This study aimed to predict the electricity consumption of elementary, middle and high schools according to the characteristics of the educational facilities in South Korea. Hong et al. [16] presented the total floor area, the building area, the total number of people, the number of students, the number of teachers, and the number of classes as influence factors in the electricity consumption of educational facilities. Table 1 shows the different characteristics of the above-mentioned six influence factors by educational facility [9]. Also, according to the Ministry of Education in South Korea, the annual average class hours in elementary, middle and high schools are 971 h/year, 1122 h/year, and 1156 h/year, respectively [43]. These characteristics by educational facility affect electricity consumption. Therefore, this study aimed to develop a prediction model by educational facility to accurately predict the trend in electricity consumption.

This study collected the monthly electricity consumption data for the last seven years (2005–2011) from a total of 787 educational facilities in Seoul (i.e., elementary school: 410; middle school: 211; and high school: 166), which was provided by the Korean Electric Power Corporation.

3.2. SARIMA model (linear approach)

The SARIMA model, as an extension of the ARIMA model which is the common linear approach for predicting future time-series,

Table 1
Characteristics by educational facility.

Classification	Total floor area (m^2/school)	Building area (m^2/school)	Total number of people (person/school)	Number of student (person/school)	Number of teachers (person/school)	Number of classes (class/school)
Elementary school	8717	2419	1014	964	50	35
Middle school	8151	2301	965	916	49	27
High school	13,755	3831	1231	1158	73	34

can improve the prediction accuracy by removing the characteristics of seasonal variation through seasonal differences. For example, electricity consumption in educational facilities is closely related to heating and cooling loads, which have seasonal characteristics in one-year interval. In this case, the SARIMA model can be implemented to remove the characteristics of seasonal variation, which can improve the prediction accuracy of future electricity consumption in educational facilities.

The SARIMA model consists of seasonal autoregressive term (P), seasonal integrated term (D), and seasonal moving average term (Q) as well as autoregressive term (p), integrated term (d), and moving average term (q). Generally, the SARIMA model represents as SARIMA (p, d, q)(P, D, Q) and is defined by equations (1)–(5). Especially, as the basic condition for the application of the SARIMA model, the mean, the variance, and the auto-correlation functions of the time series data should be stationary with time. Also, ε_t (the random error) should be independent and identically distributed as a normal distribution with zero mean. The SARIMA model is more suitable for short term than long term forecasting because it assigns higher weights to more recent data [21,28,44–48].

$$\phi_p(L)\Phi_P\left(L^S\right)(1-L)^d\left(1-L^S\right)^D Z_t = \theta_q(L)\Theta_Q\left(L^S\right)\varepsilon_t \tag{1}$$

$$\phi_p(L) = 1-\phi_1 L-\phi_2 L^2-\cdots-\phi_p L^p \tag{2}$$

$$\theta_q(L) = 1-\theta_1 L-\theta_2 L^2-\cdots-\theta_q L^q \tag{3}$$

$$\Phi_P\left(L^S\right) = 1-\Phi_S L^S-\Phi_{2S} L^{2S}-\cdots-\Phi_{PS} L^{PS} \tag{4}$$

$$\Theta_Q\left(L^S\right) = 1-\Theta_S L^S-\Theta_{2S} L^{2S}-\cdots-\Theta_{QS} L^{QS} \tag{5}$$

where, Z_t stands for the observed value at time t; ε_t stands for the lagged error at time t; L stands for the lag operator defined by $L^k Z_t = Z_{t-k}$; ϕ_p ($p = 1, 2, \cdots, p$), Φ_P ($P = 1, 2, \cdots, P$), θ_q ($q = 1, 2, \cdots, q$), Θ_Q ($Q = 1, 2, \cdots, Q$) stand for the SARIMA model parameters; p and q stand for the order of autoregressive and moving average, respectively; P and Q stand for the order of seasonal autoregressive and seasonal moving average, respectively; d and D stand for the order of difference order and seasonal difference, respectively; and S is a seasonal length.

Assuming $p = 1$, $d = 1$, $q = 1$, $P = 0$, $D = 1$, $Q = 0$ and $S = 12$, for example, SARIMA (p, d, q)(P, D, Q) can be calculated by using equations (1)–(5).

- ϕ_1 (L) and θ_1 (L) are calculated as $1-\phi_1 L^1$ and $1-\theta_1 L^1$ by using equations (2) and (3), respectively.
- Φ_1 (L^S) and Θ_1 (L^S) are calculated as $1-\Phi_1 L^{12}$ and $1-\Theta_1 L^{12}$ by using equations (4) and (5), respectively.
- The four values calculated by using equations (2)–(5) are applied to equation (1). As a result, SARIMA (1, 1, 1)(0, 1, 0) can be represented as equation (6).

$$Z_t = (Z_{t-1}+Z_{t-12}-Z_{t-13})+\phi_1(Z_{t-1}+Z_{t-2}-Z_{t-13}-Z_{t-14}) - \theta_1\varepsilon_{t-1}+\varepsilon_t \tag{6}$$

where, Z_t, Z_{t-1}, Z_{t-2}, Z_{t-12}, Z_{t-13}, and Z_{t-14} stand for the observed value at time t, $t-1$, $t-2$, $t-12$, $t-13$, and $t-14$, respectively; ε_t, and ε_{t-}

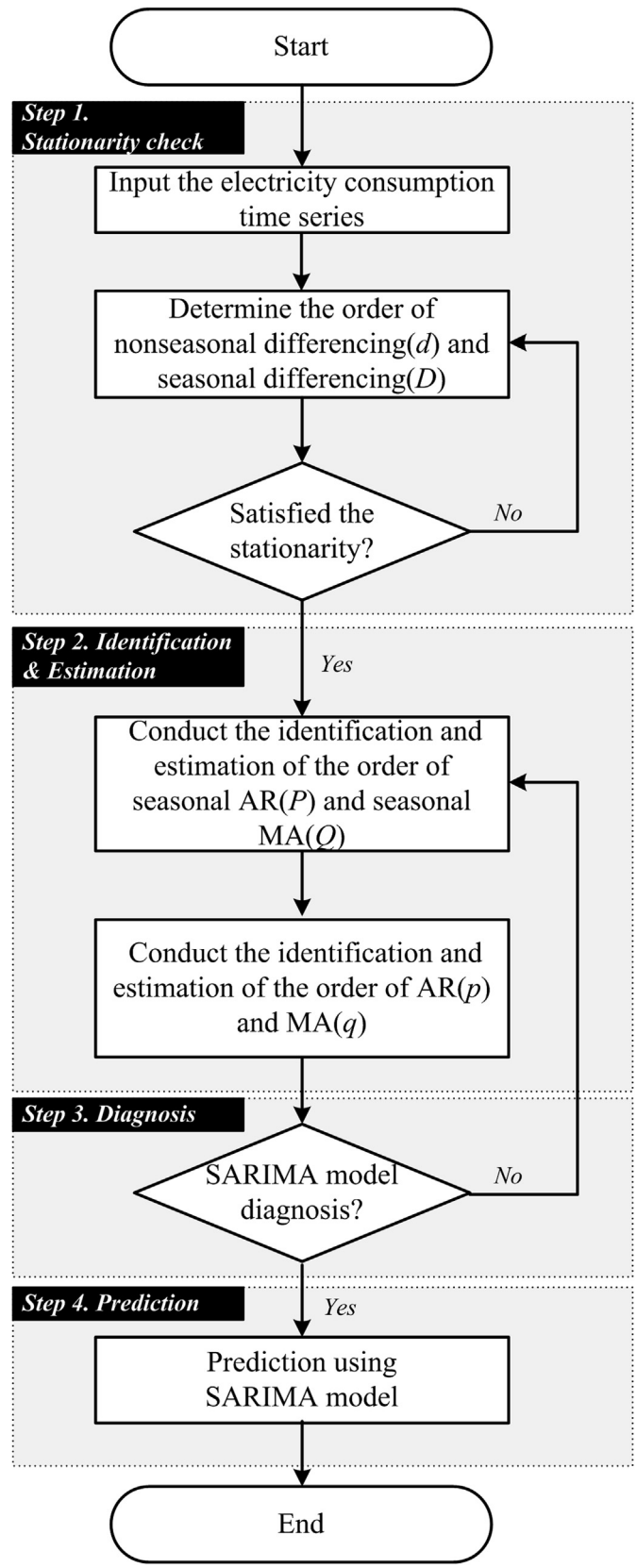


Fig. 3. The prediction process using the SARIMA model.

$_1$ stand for the random error at time t and $t-1$; and ϕ_1 and θ_1 stand for the SARIMA model parameters.

As shown in Fig. 3, the SARIMA model consists of the following four steps: (i) stationarity check; (ii) identification and estimation; (iii) diagnosis; and (iv) prediction. In step 1, a stationarity check of the time series data is conducted. Namely, the mean, the variance, and the auto-correlation functions of the time series data should be stationary with time. If time series data do not satisfy stationarity, then they can be made stationary by differencing. In step 2, identification and estimation of P (order of seasonal autoregressive), Q (order of seasonal moving average), p (order of auto-regressive), and q (order of moving average) are conducted. In step 3, the diagnosis is conducted through *Ljung–Box Q-statistics*. If p-value of *Ljung–Box Q-statistics* is larger than the 5% significance level, it can be determined that the SARIMA model is statistically appropriate. If not, step 2 (identification and estimation) should be conducted again. In step 4, time series prediction is conducted using the SARIMA model [44,49,50].

3.3. ANN model (non-linear approach)

The ANN model is a non-linear approach to solve or predict complicated problems using historically observed data [51–53]. Its basic structure is based on neurons in the human brain. Namely, just as neurons in the human brain communicate through the transmission of signals, the ANN model solves complicated problems among neurons using the combination function (linear) and activation function (non-linear) [54].

MLP (multi-layer perception) is one of the most frequently used ANN methods. As shown in Fig. 4, MLP consists of the input layer, the output layer, and more than one hidden layer. First, the observed values are combined in the input layer through the combination function (refer to equation (7)). Second, in the hidden layer, the weights of neurons are calculated through the activation function (e.g., transfer function such as the hyperbolic tangent function) (refer to equation (8)) [34,55,56]. The features of the neurons are determined by using the combination function and the activation function. Third, the optimal solution is finally represented in the output layer [26,57].

$$f(x) = \sum_i w_{il}x_i + \beta_i \tag{7}$$

where, f stands for the combination function; w_{il} stands for the weight between the ith neuron (at the input layer) and lth neuron (at the hidden layer); x_i stands for the ith neuron at the input layer; and β_i stands for the bias of ith neuron.

$$g(f(x)) = \frac{\sinh(f(x))}{\cosh(f(x))} = \frac{e^{f(x)} - e^{-f(x)}}{e^{f(x)} + e^{-f(x)}} \tag{8}$$

where, g stands for the hyperbolic tangent function as a transfer function; and f stands for the combination function.

Meanwhile, the analysis process of MLP is divided into the training phase and validation phase, which are necessary to develop the ANN model. First, in the training phase, the training data set is used to develop the optimal MLP network through an iterative learning process. Second, in the validation phase, the validation data set is applied to the optimal network established in the training phase to evaluate the prediction performance compared with the observed value [54].

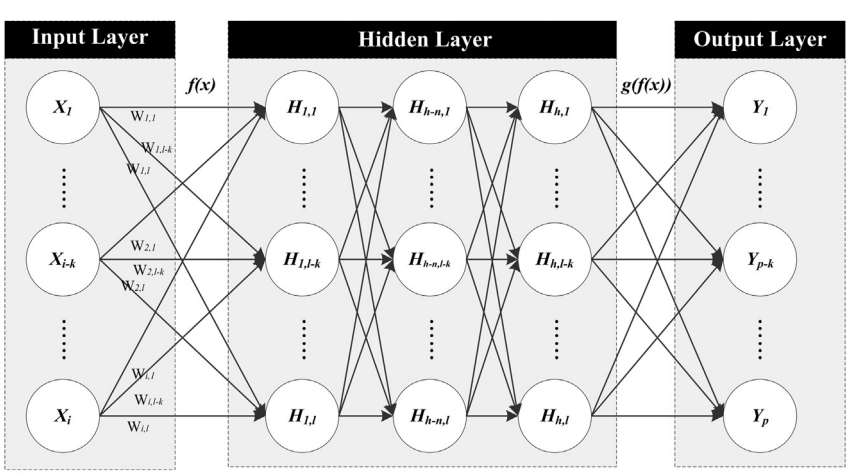
3.4. Hybrid model

The SARIMA model and the ANN model are methodologies that predict future values using historically observed data, each of which is suitable for linear and non-linear problems, respectively. The SARIMA model cannot effectively explain non-linear problems, while the ANN model offers sufficient explanatory power for non-linear problems [58]. On the other hand, the ANN model cannot effectively explain the characteristics of seasonal and trend time series [22]. As a result, if the SARIMA model or the ANN model is implemented individually, it will be difficult to solve complicated problems. Generally, however, observed values (Z_t) have both a linear factor (LF_t) and a non-linear factor (NF_t) (refer to equation (9)) [37]. Therefore, it is necessary to develop a hybrid model that combined both linear and non-linear approaches to solve this issue.

$$Z_t = f(\mathrm{LF}_t, \mathrm{NF}_t) \tag{9}$$

where, Z_t stands for the observed value at time t; LF_t stands for the linear factor; and NF_t stands for the non-linear factor.

This study developed a hybrid model that combined the SARIMA model (linear approach) and the ANN model (non-linear approach). First, the predicted value ($\widehat{L}_t$) can be calculated, as well as the predicted error (e_t^s) against the observed value (Z_t) by using the



Note. *f(x)* stands for the combination function; g(x) stands for the transfer function; X_i stands for the ith neuron at the input layer; $H_{h,n}$ stands for the nth neuron at the hth hidden layer; and Y_p stands for the pth neuron at the output layer; and n, k stands for the positive integer.

Fig. 4. The structure of ANN model.

SARIMA model (refer to equation (10)). Here, the predicted error (e_t^s) should satisfy independence and homoscedasticity [59].

$$e_t^s = Z_t - \widehat{L}_t \tag{10}$$

where, e_t^s stands for the predicted error by using the SARIMA model at time t; Z_t stands for the observed value at time t; and $\widehat{L}_t$ stands for the predicted value by using the SARIMA model at time t.

Second, the predicted error (e_t^s) that is not explained by the SARIMA model can be predicted using the ANN model (non-linear approach). The observed value (Z_{t-l}), the predicted value by using the SARIMA model ($\widehat{L}_{t-m}$), and the predicted error (e_{t-n}^s) are established as the independent variable of the ANN model. This study finally selects the model with the minimal error rate of the predicted value as the best model (refer to equation (11)).

$$\widehat{Z}_t = f_{\mathrm{ANN}}\left(Z_{t-l}, \widehat{L}_{t-m}, e_{t-n}^s\right) \tag{11}$$

where, $\widehat{Z}_t$ stands for the predicted value by using the hybrid model at time t; f_{ANN} stands for the ANN model; Z_{t-l} stands for the observed value at time $t-l$; $\widehat{L}_{t-m}$ stands for the predicted value by using the SARIMA model at time $t-m$; e_{t-n}^s stands for the predicted error by using the SARIMA model at time $t-n$; $l = 12$; $m = 0$ or 12; and $n = 0$ or 12.

4. Development of an estimation model for determining the AECB in educational facilities using SARIMA and ANN

This study collected the monthly electricity consumption data (84 points) for seven years (January 1, 2005 – December 31, 2011) by educational facility. Among them, the data for six years (January 1, 2005 – December 31, 2010) (72 points) were used in the training phase, and the other data (2011, 12 points) were used in the validation phase.

This study used the MAE (mean absolute error), RMSE (root mean square error), and MAPE (mean absolute percentage error) to assess the prediction performance of the conventional SARIMA model and the proposed hybrid models (refer to equation (12)–(14)). The smaller the MAE, RMSE, and MAPE of the prediction model are, the better its prediction accuracy is. In other words, the prediction model, which the MAE, RMSE, and MAPE are minimized, can be selected as the optimal one [36,55,60,61].

$$\mathrm{MAE} = \frac{1}{n} \times \sum_{t=1}^{n} \left| Z_t - \widehat{L}_t \right| \tag{12}$$

Table 2
Autocorrelation function of the electricity consumption by educational facility.

Classification	Autocorrelation function		
	Coefficient	Standard error	t-Statistics
Elementary school	0.686	0.244	2.81
Middle school	0.699	0.236	2.96
High school	0.675	0.212	3.18

Note. The shaded areas stand for the absolute value of t-statistics to the auto-correlation coefficient.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{t=1}^{n}\left(Z_t - \widehat{L}_t\right)^2}{n}} \tag{13}$$

$$\mathrm{MAPE} = \left(\frac{1}{n} \times \sum_{t=1}^{n} \left| \frac{Z_t - \widehat{L}_t}{Z_t} \right|\right) \times 100 \tag{14}$$

where, n stands for the total number of predictions; Z_t stands for the observed value at time t; and $\widehat{L}_t$ stands for the predicted value at time t.

4.1. Development of the SARIMA model (linear approach)

Fig. 5 shows the time series of electricity consumption by educational facility. The electricity consumptions of elementary, middle, and high schools are higher in summer and in winter. In particular, it is highest in December annually. The time series of electricity consumption tends to increase each year and have seasonality of one cycle per year. At the 5% significance level, if the absolute value of t-statistics to the auto-correlation coefficient is larger than the critical value '1.25', it can be determined that seasonality exists in the time series [44,62]. As shown in Table 2, the absolute value of t-statistics to the auto-correlation coefficient in all three types of educational facilities is larger than the critical value '1.25'. Therefore, it can be determined that seasonality exists in the time series of electricity consumption by educational facility with a time lag of 12 months.

Therefore, the SARIMA model for the time series of electricity consumption by educational facility was developed using '*IBM SPSS Statistics 21.0*'. Table 3 shows the parameter estimate results of the SARIMA model by educational facility. The SARIMA models for elementary, middle, and high schools were SARIMA (13, 1, 0)(0, 1, 0), SARIMA (6, 1, 1)(0, 1, 0), and SARIMA (6, 1, 1)(0, 1, 0), respectively. The p-values of the parameter of the SARIMA models for all three types of educational facilities were smaller than the 5% significance level, which means statistically significant.

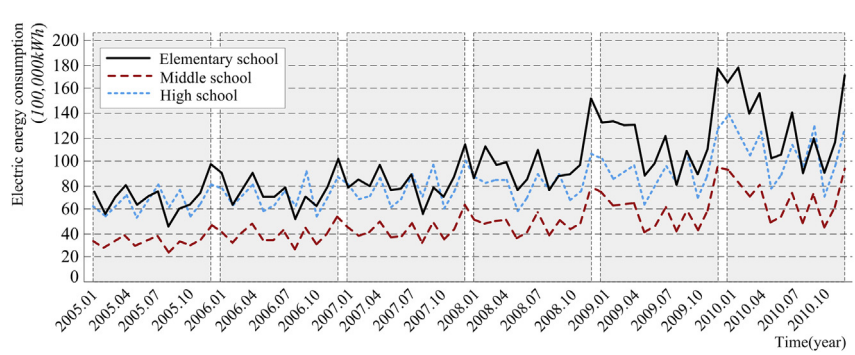


Fig. 5. Time series of electricity consumption by educational facility (2005.01–2010.12).

Table 3
Parameter estimate results of the SARIMA model by educational facility.

Classification		Lag	Estimate	SE	*t*-Statistics	*p*-Value
Elementary school	AR	13	0.391	0.147	2.655	0.010
SARIMA (13, 1, 1)(0, 1, 0)	MA	1	0.736	0.096	7.399	0.000
Middle school	AR	4	−0.273	0.13	−2.102	0.040
SARIMA (6, 1, 1)(0, 1, 0)		6	−0.292	0.133	−2.196	0.032
	MA	1	0.405	0.131	3.089	0.003
High school	AR	6	−0.298	0.134	−2.226	0.030
SARIMA (6, 1, 1)(0, 1, 0)	MA	1	0.574	0.119	4.835	0.000

AR stands for autoregressive; MA stands for the moving average; SE stands for the standard error of estimated parameters; and the shaded areas stand for the *p*-values of the parameters of the SARIMA models by educational facility.

Table 4
Ljung–Box Q-statistics by educational facility.

Classification	Ljung–Box Q-statistics		
	Q-statistics	Degree of freedom	*p*-Value
Elementary school	12.074	16	0.739
Middle school	22.171	15	0.103
High school	18.275	16	0.308

Note. The shaded areas stand for the *p*-values of the *Ljung-Box Q-statistics* of the SARIMA models by educational facility.

Finally, Table 4 shows *Ljung–Box Q-statistics* for diagnosing the SARIMA models by educational facility. The *p*-values of *Ljung–Box Q-statistics* of the SARIMA models are larger than the 5% significance level in all three types of educational facilities, which means statistically significant.

4.2. Development of the ANN model (non-linear approach)

This study developed the ANN model using '*IBM SPSS Statistics 21.0*' to predict the predicted error that has not been sufficiently explained by the SARIMA model. To improve the prediction accuracy of the ANN model, it is important to determine independent variables. This study used the following independent variables: the observed values at time $t-12$ ($Z_{t\text{-}12}$), the predicted value by using the SARIMA model at time t ($\widehat{L}_t$), the predicted value by using SARIMA model at time t-12 ($\widehat{L}_{t-12}$), the predicted error by using SARIMA model at time t (e_t^s), and the predicted error by using SARIMA model at time t-12 (e_{t-12}^s). These independent variables were established by considering that the time series data of electricity consumption by educational facility would have seasonality of one cycle per year.

5. Results and discussion

5.1. Comparison of the prediction accuracy by model

Table 5 shows the MAE, RMSE, and MAPE of the conventional SARIMA model (which only considered the linearity of the time series data) and the proposed hybrid models (which were developed by combining SARIMA and ANN) by educational facility. First, the MAE, RMSE, and MAPE of the hybrid model were lower than those of the conventional SARIMA model. Namely, it was determined that the prediction accuracy of the hybrid model was superior to that of the conventional SARIMA model. Second, the optimal hybrid models by educational facility were established depending on the different characteristics by educational facility. For elementary school, for example, hybrid model-5 ((which applied $\widehat{L}_t$ (the predicted value by using the SARIMA model at time t), e_t^s (the predicted error by using the SARIMA model at time t), and (the predicted value by using the SARIMA model at time t) to the independent variables of the ANN model)) has the highest prediction accuracy (MAE: 139,208, RMSE: 220,145, and MAPE: 0.86). Also, for middle school and high school, hybrid model-3 (MAE: 53,874, RMSE: 79,970, and MAPE: 0.72) and hybrid model-8 (MAE: 101,128, RMSE: 187,698, and MAPE: 0.82) have the highest prediction accuracy, respectively. Meanwhile, as shown in Table 5, the MAPE of the conventional SARIMA model was highest in elementary school, followed by middle school and high school. This result signifies that the nonlinearity that exists in the time series of electricity consumption by educational facility is highest in high school, followed by middle and elementary schools.

Fig. 6 shows the prediction curves for the conventional SARIMA models and the hybrid models (i.e., elementary school: hybrid model-5, middle school: hybrid model-3, and high school: hybrid model-8) by educational facility. Compared to the prediction curve of the conventional SARIMA model, the prediction curve of the hybrid model is closer to the observed value. This result was also shown identically in middle and high schools. Thus, it can be confirmed intuitively that the prediction accuracy of the hybrid model is superior to the conventional SARIMA model that only considers the linearity of the time series data.

5.2. Determination of the AECB using the hybrid model

Fig. 7 shows the MAPE of the conventional method (where the AECB in educational facilities was established based on the electricity consumption of the previous year) and the proposed method (which were developed by combining SARIMA and ANN) for determining the AECB by educational facility. First, if the conventional method was used (i.e., application of the observed electricity

Table 5
MAE, RMSE, and MAPE of the conventional SARIMA and the proposed hybrid models by educational facility.

Classification	Input modes	Elementary school			Middle school			High school		
		MAE (kWh)	RMSE (kWh)	MAPE (%)	MAE (kWh)	RMSE (kWh)	MAPE (%)	MAE (kWh)	RMSE (kWh)	MAPE (%)
Conventional SARIMA model	–	1,009,367	1,147,899	8.52	741,811	845,981	11.13	1,396,484	1,558,306	14.13
Hybrid model-1	$\widehat{L}_t, e_t^s, \widehat{L}_{t-12}, e_{t-12}^s, Z_{t-12}$	189,461	289,261	1.22	127,001	167,644	1.69	160,845	255,001	1.36
Hybrid model-2	$\widehat{L}_t, e_t^s, \widehat{L}_{t-12}, e_{t-12}^s$	156,670	250,496	0.98	80,976	135,838	1.02	123,939	212,145	1.03
Hybrid model-3	$\widehat{L}_t, e_t^s, \widehat{L}_{t-12}, Z_{t-12}$	158,020	252,594	0.97	53,574	79,970	0.72	149,296	211,269	1.31
Hybrid model-4	$\widehat{L}_t, e_t^s, e_{t-12}^s, Z_{t-12}$	191,281	310,992	1.22	186,868	218,207	2.76	140,033	238,931	1.15
Hybrid model-5	$\widehat{L}_t, e_t^s, \widehat{L}_{t-12}$	139,208	220,145	0.86	76,688	99,074	1.06	145,104	205,851	1.19
Hybrid model-6	$\widehat{L}_t, e_t^s, e_{t-12}^s$	246,554	398,712	1.56	120,050	215,402	1.48	127,821	206,302	1.07
Hybrid model-7	$\widehat{L}_t, e_t^s, Z_{t-12}$	158,824	265,753	0.95	75,680	106,104	1.05	119,457	196,532	0.99
Hybrid model-8	$\widehat{L}_t, e_t^s$	170,325	256,993	1.12	78,377	119,071	1.10	101,128	187,698	0.82

$\widehat{L}_t$ stands for the predicted value by using the SARIMA model at time t; e_t^s stands for the predicted error at time t by using SARIMA model; $\widehat{L}_{t-12}$ stands for the predicted value by using the SARIMA model at time t; e_{t-12}^s stands for the predicted error at time $t-12$; $Z_{t\text{-}12}$ stands for the observed value at time $t-12$; and the shaded areas stand for the optimal hybrid model by educational facility.

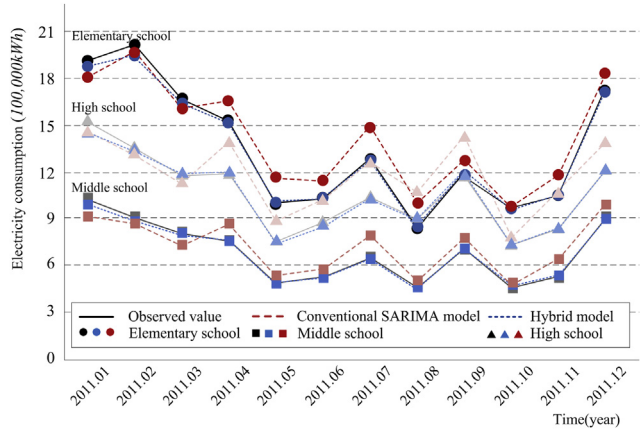
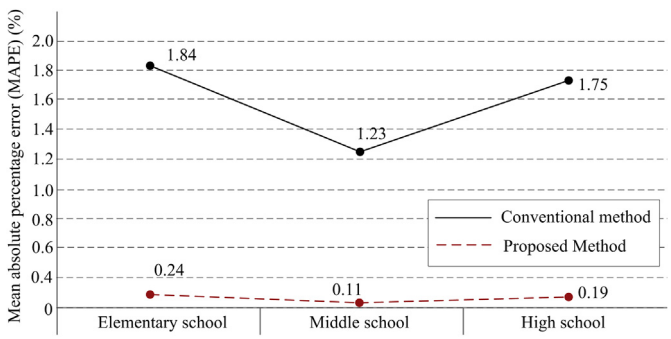


Fig. 6. Prediction curves by educational facility using the conventional SARIMA model and hybrid model.



Note. MAPE (A) stands for the mean absolute percentage error between the observed value in the previous year (2010) and the observed value in the current year (2011); and MAPE (B) stands for the mean absolute percentage error between the predicted value by using the hybrid model-8 and the observed value in the current year (2011).

Fig. 7. MAPE of the conventional method and the proposed method for determining the AECB by educational facility.

consumption in the previous year (2010)), the MAPEs for elementary, middle, and high schools were 1.84%, 1.23%, and 1.75%, respectively. Second, if the proposed method was used (i.e., application of the predicted electricity consumption using the hybrid model), the MAPEs for all three types of educational facilities were less than 0.3% (i.e., elementary school: 0.24%, middle school: 0.11%, and high school: 0.19%). Thus, it has been proved that the proposed method enables executives and managers in charge of budget planning to establish the AECB in educational facilities more accurately, compared to the conventional method.

Currently, the AECB of educational facilities is established based on the electricity consumption of the previous year. However, an inaccurate AECB plan has caused both insufficient operation of heating and cooling systems, degrading the quality of the learning environment and the lack of a budget for the other purposes, resulting in the unbalanced budget for the building operation and maintenance. If the AECB can be more accurately established based on the proposed method, sufficient operation of heating and cooling systems and balanced execution of the budget will be possible, which will ultimately improve the quality of the learning environment.

6. Conclusions

This study aimed to develop an estimation model for determining the AECB in educational facilities using SARIMA and ANN. This study was conducted in three steps: (i) establishment of electricity consumption data for educational facilities; (ii) estimation of the annual electricity consumption in educational facilities using the SARIMA model; and (iii) improvement of prediction accuracy by considering the non-linearity of the residual from the SARIMA model using the ANN model.

When the hybrid models developed in this study were used, the MAE, RMSE, and MAPE of the hybrid models were lower than those of the conventional SARIMA models that only consider the linearity of the time series data. Namely, for elementary, middle, and high schools, the hybrid model-5 (MAE: 139,208, RMSE: 220,145, and MAPE: 0.86), hybrid model-3 (MAE: 53,874, RMSE: 79,970, and MAPE: 0.72), and hybrid model-8 (MAE: 101,128, RMSE: 187,698, and MAPE: 0.82) have the highest prediction accuracy, respectively. Meanwhile, it was determined that the proposed method for determining the AECB in educational facilities was superior to the conventional method. If the conventional method was used, the MAPEs for elementary, middle and high schools were 1.84%, 1.23%, and 1.75%, respectively. If the proposed method was used, it was shown as below 0.3% in all three types of educational facilities (i.e., elementary school: 0.24%, middle school: 0.11%, and high school: 0.19%).

The proposed model could enable executives and managers in charge of budget planning to accurately estimate the annual energy consumption and determine the AECB in educational facilities. It could be also applied to other types of resources (e.g., water consumption or gas consumption) in educational facilities. Furthermore, other methods, such as support vector machines or case-based reasoning, could be used in future studies to develop a hybrid model to improve prediction accuracy.

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